Transformations from Non-Abelian to Abelian Galois Groups in Hamiltonian Dynamics: A Classification and Synthesis

I. The Galoisian Obstruction: A Precise Formulation of the Morales-Ramis-Simó Criterion

The central difficulty in Hamiltonian dynamics is rigorously distinguishing between integrable and chaotic (non-integrable) systems.[1] The Morales-Ramis theory provides the most potent algebraic criterion for this, translating the analytic problem of integrability into a group-theoretic problem within differential Galois theory.[3] This section will precisely formulate this obstruction and establish the non-integrable baseline (the three-body problem) from which the query's transformations will begin.

A. The Fundamental Theorem: Liouville Integrability and Abelian G°

The concept of complete integrability for a Hamiltonian system X_H on a complex analytic symplectic manifold M of dimension $2n$ is defined in the Liouville sense: the system must admit n functionally independent meromorphic first integrals that are in involution (i.e., their Poisson brackets mutually vanish).[4]

The Morales-Ramis theorem establishes a profound, necessary condition for this. It connects the analytic properties of the system (integrability) to the algebraic structure of its linear approximation along a trajectory.[9]

Let $\phi(t)$ be a non-equilibrium particular solution of X_H, which defines a complex analytic curve (a Riemann surface) Γ.4 The Variational Equation (VE) along

Γ is the linear differential equation governing the evolution of an infinitesimal displacement $y \in T_\Gamma M$:

$$\dot{y} = A(t)y, \quad \text{where} \quad A(t) = \frac{\partial X_H}{\partial x}(\phi(t))$$

The coefficients of the matrix $A(t)$ belong to the field of functions meromorphic on Γ, denoted $M(\Gamma)$.4 This field, equipped with the derivative $\frac{d}{dt}$, is a differential field.10
The **Differential Galois Group (DGG)**, G, of the VE is the group of differential automorphisms of the Picard-Vessiot extension (the smallest differential field extension of $M(\Gamma)$ containing all solutions to the VE).[3] G is a linear algebraic group and, because the VE is Hamiltonian, G is an algebraic subgroup of the symplectic group $Sp(2n, \mathbb{C})$.[4]

The Morales-Ramis Theorem states 4:
If a Hamiltonian system is meromorphically integrable in the Liouville sense in a connected neighborhood U of the phase curve Γ, then the identity component (G°) of the differential Galois group G of the Variational Equation along Γ must be Abelian.
The identity component G° is the largest connected algebraic subgroup of G containing the identity.[13] The obstruction to integrability is the contrapositive: if one can compute the DGG for a particular VE and finds that its identity component G° is *non-abelian*, the system is proven to be meromorphically non-integrable in the Liouville sense.[5]

A crucial refinement of the hypotheses, noted in [18], is that the n first integrals must be meromorphic, in involution, and functionally independent in the neighborhood U, but they are *not* required to be independent on the curve Γ itself. This makes the obstruction stronger, as it applies to a wider class of solutions.[18]

B. Essential Distinctions and Generalizations

The applicability of the Galoisian criterion depends on several crucial distinctions.

VE vs. NVE (Normal Variational Equation)

The full VE of rank $2n$ always possesses at least one first integral: the differential of the Hamiltonian, $dH(\phi(t))$, which is an integral of the VE.[9] Using a D'Alambert reduction, this allows the system to be reduced to the **Normal Variational Equation (NVE)**, which is a linear symplectic system of rank $2(n-1)$.[19] For Liouville integrability, the identity components of the DGGs of *both* the full VE and the NVE must be abelian.[18] Proving that the G° of the NVE is non-abelian is sufficient to prove the non-integrability of the full system and is often computationally simpler.[18]

Liouville Integrability vs. Integrability by Quadratures

The user's query specifies an "abelian identity component," which is the criterion for Liouville integrability. However, a related and weaker concept is *integrability by quadratures*, which means the solution field of the VE is a *Liouvillian extension* (it can be constructed using a tower of integrals, exponentials of integrals, and algebraic functions).[3]

These two concepts correspond to different algebraic structures for G°:

1. **Liouville Integrability (of H) $\implies G^\circ(\text{VE})$ is Abelian**.[4]
2. **Integrability by Quadratures (of VE) $\iff G^\circ(\text{VE})$ is Solvable**.[3]

Since any abelian group is solvable, the Liouville condition is strictly stronger. A transformation that "simplifies" a non-abelian, non-solvable G° (e.g., one containing $SL(2, \mathbb{C})$) to a *solvable but non-abelian* G° would not make the Hamiltonian Liouville integrable, but it *would* make its variational equation solvable by quadratures. This report will classify transformations that can achieve either the solvable (weaker) or abelian (stronger) target.

Morales-Ramis-Simó (Higher Order VEs)

The Galoisian obstruction is, in fact, much stronger. As shown by Morales, Ramis, and Simó, for complete meromorphic integrability, the identity component G° of the DGG of the variational equations of *all orders* ($k \ge 1$) must be abelian.[25] It is possible for a system to pass the first-order test ($G^\circ(\text{VE}_1)$ is abelian) but fail at the second order ($G^\circ(\text{VE}_2)$ is non-abelian), thus proving its non-integrability.[25]

B-integrability (Bogoyavlenskij-integrability)

The query also mentions transforming the system to a **B-integrable** one. This concept, introduced by Bogoyavlenskij, typically applies to non-Hamiltonian systems.[29] The literature asserts that a "beautiful generalization of the Morales-Ramis Theorem" exists for systems that are B-integrable.[10] However, the precise nature of this generalized Galoisian constraint (i.e., what property G° must satisfy) is not specified in the available reference materials.[10] This constitutes a significant gap in the literature. Without a well-defined algebraic target for G°, it is impossible to definitively confirm if a transformation has successfully produced a B-integrable system *via the Galoisian criterion*. Therefore, this analysis will focus on the well-defined criteria for Liouville integrability (abelian G°) and integrability by quadratures (solvable G°).

C. The Non-Abelian Baseline: Three-Body Orbits

The query's premise—that certain three-body orbits provide a non-abelian baseline—is robustly supported by the literature.

- **Planar 3-Body Problem:** The non-integrability of the planar three-body problem in the neighborhood of the Lagrangian (equilateral triangle) solutions was established by Tsygvintsev [31] and Maciejewski & Przybylska.[4] Tsygvintsev's original work used the monodromy group (a dense subgroup of the DGG [34]) along parabolic orbits to demonstrate a lack of rational invariants, proving non-integrability.[33]
- **Planar N-Body Problem:** Maciejewski & Przybylska later proved the non-integrability of the general planar n-body problem when restricted to a common level of energy and angular momentum.[35]
- **Charged 3-Body Problem:** For the charged three-body problem (with Coulomb or gravitational potentials), non-integrability is proven if at least two of the interaction parameters are non-zero.[15] The proof is a canonical application of the Morales-Ramis theorem: one finds a Darboux point (an equilibrium of the potential on the unit sphere) and computes the eigenvalues of the Hessian, showing they violate the necessary conditions required for G° to be abelian.[17]
- **Restricted 3-Body Problem (R3BP):** The non-integrability of the R3BP has been proven using the higher-order Morales-Ramis-Simó theory [28] and generalized versions of the theory (e.g., Ayoul-Zung).[16]

Collectively, these results establish the 3-body problem as a canonical source of VEs and NVEs whose DGG identity components G° are known to be non-abelian (and often

non-solvable, containing $SL(2, \mathbb{C})$ or $Sp(2n, \mathbb{C})$). These systems serve as the non-integrable "input" for the transformations this report will now classify.

Table 1: Galoisian Constraints for Notions of Integrability

Integrability Type	Definition	DGG (G°) Constraint	Reference(s)
Liouville Integrability	n meromorphic, independent, commuting first integrals.	**Abelian**	[4]
Integrability by Quadratures	Solution field is a Liouvillian extension.	**Solvable**	[3]
B-integrability	Bogoyavlenskij integrability for (non-)Hamiltonian systems.	****	[10]

II. Invariance of the Obstruction: Transformations that Preserve Non-Integrability

Before classifying effective transformations, it is essential to identify those that *fail* to alter the Galoisian obstruction. These transformations preserve the non-abelian nature of G° and are thus ineffective for the query's goal.

A. Symplectic Coordinate Changes (Meromorphic on Γ)

A natural first attempt is to change coordinates on the phase space M via a (meromorphic) symplectic map Φ.

- **Action:** A VE $\dot{y} = A(t)y$ defined over the differential field $k = M(\Gamma)$ is transformed into a new system $\dot{z} = B(t)z$. The transformation matrix $P(t) = d\Phi(\phi(t))$ is symplectic and has entries in the base field k.
- **Effect on VE:** The new VE matrix is given by $B(t) = P(t)^{-1}A(t)P(t) - P(t)^{-1}\dot{P}(t)$.[43]
- **Effect on DGG:** This transformation is a **gauge equivalence** over the base field k.[43] The differential Galois group is an invariant of the underlying *differential module* (an abstract representation of the VE as a vector space with a connection).[44] As such, the DGG is, by definition, invariant under gauge transformations.
- **Conclusion:** Meromorphic symplectic coordinate changes are **ineffective**. They merely change the basis in which the differential module is expressed, but they do not alter its intrinsic algebraic structure. The non-abelian G° is preserved.

B. Time Reparametrizations (Algebraic)

Another "obvious" transformation is a general change of the independent variable (time), $t = f(\tau)$.

- **Action:** The VE $\frac{dy}{dt} = A(t)y$ becomes $\frac{dy}{d\tau} = \left(A(f(\tau)) \cdot f'(\tau) \right) y = B(\tau)y$.
- **Effect on DGG:** This transformation *does* change the base differential field, from $k = M(\Gamma_t)$ to $k' = M(\Gamma_\tau)$. However, the *solution field* L (the Picard-Vessiot extension) remains the same, as the solutions y are the same analytic functions, merely composed with $f(\tau)$. The DGG is defined as $G = \text{Gal}(L/k)$ or $G' = \text{Gal}(L/k')$.
- **Invariance of the Criterion:** A fundamental physical and mathematical requirement is that the *criterion* for non-integrability must be invariant under a mere reparametrization of the solution's trajectory.[45] The algebraic property of G° (being abelian or solvable) is an intrinsic property of the *system* (the geometry of its solution $\phi(t)$), not the *speed* at which the solution is traversed. The Ziglin-Morales-Ramis analysis is fundamentally about the branching (monodromy) of solutions in the complex plane, a property that is not removed by a simple reparametrization.
- **Conclusion:** General time reparametrizations are **ineffective**. They do not alter the abelian/solvable nature of G°. (This must be sharply distinguished from the *specific* case of *ramified coverings*, which are field extensions, as detailed in Section

III.A).

III. "Abelianization" via Field Extensions: Coverings and the Inverse Problem

This section details the first class of *effective* transformations: those that directly modify the algebraic structure of the problem by extending the base differential field over which the VE is defined.

A. Algebraic Coverings of the Independent Variable

This is a specific, powerful type of time change that *does* alter the DGG.

- **Action:** A ramified covering of the independent variable, such as $t = s^m$ for an integer $m > 1$.[19]
- **Effect on VE:** The VE $\frac{dy}{dt} = A(t)y$, defined over the base field $k = \mathbb{C}(t)$ (assuming an algebraic curve Γ), is transformed into $\frac{dy}{ds} = (m s^{m-1} A(s^m)) y$, which is now defined over the *new* base field $k_s = \mathbb{C}(s)$.
- **Effect on DGG:** The field k_s is a finite algebraic extension of k. Let L be the original Picard-Vessiot (PV) extension for the VE over k. The new PV-extension over k_s is the compositum field $L_s = L \cdot k_s$. By the fundamental theorem of differential Galois theory, the new DGG $G_s = \text{Gal}(L_s / k_s)$ is an algebraic subgroup of the original DGG $G = \text{Gal}(L/k)$.
- **Mechanism of "Abelianization":** The original G° may be non-abelian, but it is possible for its algebraic subgroup G_s° to be abelian (or solvable). This is a standard technique for simplifying VEs. For example, a VE with an irregular singularity at $t=\infty$ (a common source of non-abelian DGGs [19]) can sometimes be "unwound" by a covering $t=s^m$ into a new VE over k_s that has only regular singularities. This process, known as algebrization, simplifies the local monodromy and can reduce the DGG.
- **Conclusion:** This is a **Class 1 (Algebraic)** transformation. It *directly reduces* the DGG by extending the base field. The challenge is non-trivial: one must find the specific covering m that induces the desired simplification of the group structure (e.g., from $SL(2, \mathbb{C})$ to a solvable or abelian subgroup).

B. The Inverse Differential Galois Problem (IDGP)

The IDGP provides the constructive, albeit non-dynamical, method for "forcing" an abelian group.

- **The Problem:** The IDGP asks: given a linear algebraic group G (e.g., an abelian group), can one *construct* a linear differential equation $L(y)=0$ over $\mathbb{C}(t)$ whose DGG is G?.[46]
- **Relevance:** The literature provides an affirmative answer and constructive methods for the groups in question.
    - **Solvable Groups:** Shafarevich proved that *all* finite solvable groups are realizable as Galois groups over $\mathbb{Q}$.[48]
    - **Abelian Groups:** All finite abelian groups are realizable, often via cyclotomic field extensions.[50]
    - **Constructive Methods:** The work of Singer, van der Put, Mitschi, and others provides algorithms and frameworks for constructing linear differential equations with prescribed solvable or abelian DGGs.[46]
- **Stitching the Gap (The "Transformation"):** The query asks how to *transform* the given non-abelian 3-body VE, $L_1(y)=0$, into a new system $L_2(y)=0$ with an abelian DGG. The IDGP literature does not provide a transformation $L_1 \to L_2$. It provides the *existence* and *construction* of L_2 from scratch.
  The "transformation" is thus the abstract algebraic act of **field extension**.
    1. Start with $L_1(y)=0$ (the 3-body VE) over $k = M(\Gamma)$, with non-abelian DGG $G_1 = \text{Gal}(L_1/k)$.
    2. The target is a system $L_2(y)=0$ with an abelian DGG G_2. By definition, the solution field L_2 for such a system is a Liouvillian extension of k.[23]
    3. We can "force" the original problem to become abelian by defining a new base field $k' = k \cdot L_2$ (the compositum of k and this new Liouvillian extension).
    4. The DGG of the *original* equation $L_1(y)=0$, when considered over this *new, larger* base field k', is $G' = \text{Gal}(L_1 \cdot k' / k')$.
    5. By Galois correspondence, G' is an algebraic subgroup of G_1. By adjoining a sufficiently large Liouvillian extension L_2, the DGG of the original VE can be made trivial (if $L_1 \subset k'$) or, more generally, abelian.
- **Conclusion:** This is an algebraic, non-dynamical "forcing" of the abelian group. It does not create a new *dynamical* system in the sense of the query, but it provides the underlying algebraic mechanism for how a DGG can be reduced.

IV. "Abelianization" via Geometrization: Lifts and Enlarged Phase Spaces

This section details the most sophisticated class of transformations. Instead of simplifying the DGG algebraically, these methods *re-frame* the entire dynamical problem by lifting it into a new, larger phase space. In this new space, the original algebraic obstruction (non-abelian G°) is mapped to a different, often geometric, criterion.

A. The Eisenhart-Duval Lift

This transformation provides the most direct link between the DGG of a Hamiltonian system and the geometric properties of a lifted space.

- **Action:** The Eisenhart-Duval lift geometrizes a classical system $H = T + V$ (on a configuration space M) by mapping its trajectories to **null geodesics** on a larger, Lorentzian manifold (M_E, G) of dimension $\text{dim}(M)+2$.[56]
- **The Critical Equivalence:** A direct link between the Liouville integrability of H (and thus its abelian G°) and the symmetries of M_E is established in.[58] In a study of a quantum dot potential, it was shown that:
    1. The Hamiltonian H is Liouville integrable if and only if a specific parameter τ^2 belongs to the set $\{1, 4, 1/4\}$.[59]
    2. This integrability was proven using the Morales-Ramis theorem, explicitly stating that these are the cases where the G° of the VE is **Abelian**.[59]
    3. This condition ($\tau^2 \in \{1, 4, 1/4\}$) was proven to be *equivalent* to the existence of an **irreducible conformal Killing tensor** on the associated Eisenhart manifold M_E.[59]
- **Mechanism of "Abelianization":** This equivalence provides a "Rosetta Stone" for this part of the query.
    1. The starting 3-body system has a *non-abelian G°*.
    2. By this equivalence, its Eisenhart lift M_E must *lack* the required Killing tensors. The algebraic obstruction (G°) in the base space *is* the geometric obstruction (lack of Killing tensors) in the lifted space.
    3. The "transformation" is the lift itself. The "new system" is the *geodesic flow* on M_E. The "new variational equation" is the *Jacobi equation* (geodesic deviation equation) for that null geodesic.
    4. Therefore, the Eisenhart lift transforms the problem: the search for an abelian G° becomes the search for a Killing tensor. The transformation "succeeds"

by mapping the purely algebraic problem into an equivalent problem in Riemannian geometry.

- **Conclusion:** This is a **Class 2 (Geometric - Equivalence)** transformation.

B. The Jacobi-Maupertuis (JM) Principle

This transformation also geometrizes the dynamics but with a fundamentally different effect on the variational equations.

- **Action:** The JM principle transforms the Hamiltonian flow on a constant energy surface $H=E$ into a **geodesic flow** on the original configuration manifold M, but equipped with a new Riemannian metric known as the Jacobi metric: $g_{JM} = (E-V)g_0$ (where V is the potential and g_0 is the kinetic energy metric).[60]
- **The Critical Dis-analogy:** The original VE (linearization of the time-dependent flow) and the new VE (the *Jacobi equation*, or geodesic deviation equation, for the g_{JM} geodesic flow [63]) are **not equivalent**.[65] They are distinct linear systems.
- **Integrability of the Lift:** The *new* system (the geodesic flow) is itself a Hamiltonian system, and its integrability can be analyzed *independently* using the Morales-Ramis theorem.[67] The complete integrability of the geodesic flow (e.g.[68]) depends on whether the G° of the *Jacobi equation* is abelian.
- **Mechanism of "Abelianization":**
  1. The starting 3-body system has a non-abelian $G^\circ(\text{VE}_H)$.
  2. The JM lift *discards* this VE_H and its associated DGG entirely.
  3. It creates a *new* dynamical system (geodesic flow) with a *new* variational equation (the Jacobi eq., VE_{JM}) and a *new*, *independent* DGG, $G(\text{VE}_{JM})$.
  4. This transformation "succeeds" if the new system (the geodesic flow) *happens* to be integrable (i.e., its *own* $G^\circ(\text{VE}_{JM})$ is abelian), even if the original time-dependent system was not.
- **Conclusion:** This is a **Class 3 (Geometric - Parallel)** transformation. It does not *transform* the original DGG; it *replaces* the original dynamical problem with a parallel, non-equivalent one.

C. Regularization Lifts (Kustaanheimo-Stiefel)

This class of transformations addresses the source of non-integrability (singularities) directly.

- **Action:** The Kustaanheimo-Stiefel (KS) transformation is a canonical lift from the 3D physical space to a 4D "oscillator" space, combined with a *simultaneous time reparametrization* $dt = r \cdot ds$, where r is the distance to the singularity.[70]
- **Effect:** Its primary purpose is *regularization*. It famously maps the *singular* 2-body Kepler problem into the *regular, linear* 4D harmonic oscillator.[73] In the 3-body problem, it is used to remove the collision singularities that are the source of chaotic behavior.[75]
- **The Critical Insight:** The 2D version of this transformation (the Levi-Civita regularization) is shown to produce a *local first integral* for the restricted three-body problem in a neighborhood of the regularized collision singularity.[75]
- **Mechanism of "Abelianization":**
    1. The non-integrability of the 3-body problem, and thus its non-abelian G°, is intrinsically linked to the complex branching of solutions around its *singularities* (which are often irregular singularities of the VE [19]).
    2. The KS transformation is explicitly designed to *remove* these singularities.[76]
    3. The existence of a *local first integral* in the *regularized variables* [75] implies, by the Morales-Ramis theorem, that the VE of the *regularized Hamiltonian* [79] must have a **local abelian G°**.
    4. This transformation does not "abelianize" the *global* DGG of the original problem. It creates a new, locally-defined system where the *source* of the non-abelian behavior (the singularity) has been removed.
- **Conclusion:** This is a **Class 4 (Geometric - Local)** transformation. It achieves *local* Liouville integrability (and thus a local abelian G°) by lifting and reparametrizing to excise the singularities.

V. Synthesis and Gaps: Towards an "Inverse Morales-Ramis" Framework

This analysis culminates in a classification of the transformations capable of mapping a system with a non-abelian G° to one with an abelian (or solvable) G°. These transformations fall into four distinct categories, summarized below.

A. Classification of "Abelianizing" Transformations

The transformations can be categorized by their fundamental mechanism:

- **Class 0 (Invariant):** Transformations that *preserve* the non-abelian G°.
- **Class 1 (Algebraic):** Transformations that *reduce* the DGG by extending the base field.
- **Class 2 (Geometric - Equivalence):** Transformations that *re-frame* the algebraic obstruction as an equivalent geometric one.
- **Class 3 (Geometric - Parallel):** Transformations that *replace* the original problem with a new, non-equivalent one.
- **Class 4 (Geometric - Local):** Transformations that *remove the source* of the non-abelian G° (singularities) to achieve local integrability.

Table 2: Classification of Transformations and their Effect on the Galoisian Obstruction

Transformation Class	Specific Example	Mechanism of Action on DGG G°	Effectiveness for Abelianization
Class 0: Invariant	1. Meromorphic Symplectic Coord. Change 2. Algebraic Time Reparametrization	1. **Gauge Equivalence** over the base field. G° is a DGG invariant.[43] 2. **Invariance of the Criterion**. The DGG G and its property of being (non-)abelian is an intrinsic property of the system, not its parametrization.[45]	**Ineffective.** Preserves the non-abelian G°.
Class 1: Algebraic	Algebraic Covering (e.g., $t = s^m$)	**Field Extension (Subgroup).** Creates a new VE over an extended field $k_s \supset k$. The new DGG G_s° is an	**Effective (Direct).** A non-abelian G° can have an abelian subgroup G_s°.

		algebraic subgroup of the original G°.[19]	
Class 2: Geometric (Equivalence)	Eisenhart-Duval Lift	**Re-framing (Equivalence).** Lifts H to a null geodesic flow. Maps the algebraic obstruction (non-abelian G°) to an *equivalent geometric* one (absence of Killing tensors).[58]	**Effective (Conceptual).** "Solves" the problem by mapping it to an equivalent problem in geometry.
Class 3: Geometric (Parallel)	Jacobi-Maupertuis (JM) Lift	**Replacement (Parallel).** Discards the original VE. Creates a *new* VE (the Jacobi Eq.) for a *new* system (geodesic flow). The new $G^\circ(\text{VE}_{JM})$ is *not equivalent* to the old $G^\circ(\text{VE}_H)$.[65]	**Effective (Indirect).** Succeeds if the *new* system is integrable, regardless of the original.
Class 4: Geometric (Local)	Kustaanheimo-Stiefel (KS) Lift	**Singularity Removal (Local).** Lifts coordinates and reparametrizes time ($dt=r ds$) to remove singularities.[75]	**Effective (Local).** Achieves *local* Liouville integrability (and thus local abelian G°) in a neighborhood of the regularized collision.[75]

B. Identified Gaps and Future Research

This "stitching" of results reveals several critical gaps in the existing literature, highlighting the open nature of a true "Inverse Morales-Ramis" theory.

1. **Gap 1: The B-integrability Criterion.** The most significant gap identified. The literature *claims* a "beautiful generalization" of the Morales-Ramis theorem exists for B-integrability [10], but its precise algebraic constraint on G° (e.g., solvable? abelian? virtually solvable?) is not specified. This makes it impossible to definitively classify or verify transformations to *achieve* this state using the DGG.
2. **Gap 2: The DGG(VE) $\to$ DGG(JM) Mapping.** The JM lift (Class 3) is a "black box" in this context. It is known that the VE of the Hamiltonian flow and the Jacobi equation of the geodesic flow are *not* equivalent.[65] However, there is no known *predictive* theory that maps the properties of the original $G^\circ(\text{VE}_H)$ to the resulting $G^\circ(\text{VE}_{JM})$. The literature [67] simply applies the Morales-Ramis theorem *anew* to the resulting geodesic flow, without reference to the original DGG.
3. **Gap 3: The "Stitching" Problem.** The user's query implicitly seeks a unified "Inverse Morales-Ramis Theorem," which does not exist. The literature is fragmented into the four classes identified:
    - **Class 1 (Algebraic)** is constructive [46] but non-dynamical.
    - **Classes 2-4 (Geometric)** are dynamical but are not simple "simplifications" of the DGG; they are deep, conceptual *reframings* of the entire integrability problem.

The synthesis of these approaches remains an open and advanced research problem. For example: can a **Class 4 (KS)** lift be applied to a 3-body orbit to *regularize* it, followed by a **Class 2 (Eisenhart)** lift of the *regularized* Hamiltonian, to then study its global integrability via the existence of Killing tensors? This "stitching" of transformations to navigate between algebraic and geometric criteria for integrability is the frontier of this field.

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